Cubic Regularization Methods

Theory, Adaptivity, and Empirical Evaluation

[Team Name]

Continuous Optimization — Project Presentation

April 3, 2026

Outline

Introduction & Motivation

Related Work

Methodology & Preliminary Results

Section 1

Introduction & Motivation

Nonconvex Optimization is Everywhere

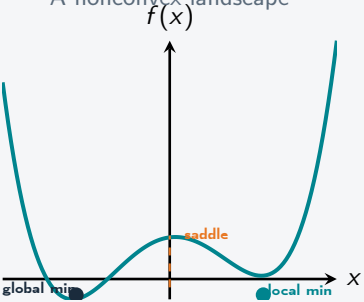
The challenge: many real problems are nonconvex

- Training neural networks (non-convex loss landscapes)
- Logistic regression with non-convex regularizers
- Scientific computing & engineering design
- Statistical learning on non-standard loss functions

What do we want from an optimizer?

- **Global guarantees** — convergence from any starting point
- **Fast local behaviour** — exploit curvature near solutions
- **Provable complexity** — worst-case iteration bounds

A nonconvex landscape



Why Newton's Method is Not Enough

Newton's update:
$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$

Three fundamental failures

1. **Hessian indefinite** $\Rightarrow$ step may *ascend*
2. **No global convergence** guarantee — can diverge or cycle
3. **Saddle attraction** — converges to non-minimizers

Worst-case complexity: $\boxed{\mathcal{O}(\varepsilon^{-2})}$ — no better than gradient descent!

The key insight: if $\nabla^2 f$ is *Lipschitz* with constant L,

Taylor remainder bound

$$f(x_k+s) \leq \underbrace{f(x_k) + \nabla f(x_k)^\top s + \tfrac{1}{2} s^\top \nabla^2 f(x_k)\, s}_{\text{quadratic model}} +$$

$\Rightarrow$ **Minimizing this cubic upper bound guarantees descent** even with an indefinite Hessian!

Project Goals

Objectives

1. **Theory** — study the mathematical foundations and historical development of cubic regularization methods
2. **Comparison** — analyse CRN, ARC vs. gradient descent, conjugate gradient, quasi-Newton
3. **Complexity** — examine worst-case iteration bounds and their optimality
4. **Experiments** — implement selected methods in Python and evaluate on two real datasets

Datasets (planned):

- **Covtype** — forest cover classification (~581k samples)
- **A9a** — census binary classification (32k samples)

Loss functions:

- Logistic regression (convex)
- Logistic + non-convex regularizer [Reddi et al. 2016]

Section 2

Related Work

Historical Development

Griewank
First cubic model for Newton's method;
global convergence to 2nd-order critical pts.

Nesterov
Accelerated cubic for convex problems;
$\mathcal{O}(1/k^3)$ function-value rate.

Stochastic variants,
lower bounds, Riemannian,
sub-sampled methods.

1981 — 2006 — 2008 — 2011 — 2017+

Nesterov & Polyak
Formal analysis;
first $\mathcal{O}(\varepsilon^{-3/2})$ complexity bound.

Cartis, Gould & Toint
ARC: adaptive σ_k, approximate Hessians, practical complexity.

The Cubic Regularized Model

At iterate x_k, define the **cubic model**:

$$\boxed{m_k(s) = f(x_k) + g_k^\top s + \tfrac{1}{2} s^\top B_k s + \frac{\sigma_k}{3} \|s\|^3}$$

where $g_k = \nabla f(x_k)$, $B_k \approx \nabla^2 f(x_k)$, $\sigma_k > 0$.

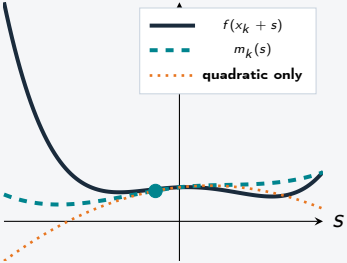
First-order optimality condition for s^*

$$(B_k + \lambda^* I)\, s^* = -g_k, \quad \lambda^* = \sigma_k \|s^*\|$$

$$B_k + \lambda^* I \succeq 0$$

Analogous to trust-region, but λ^* is determined endogenously by the cubic penalty.

Model vs. true function



The cubic model is a *global upper bound* when $\sigma_k \geq L/2$.

Model Decrease and Complexity

Model decrease bound (from optimality conditions):

$$f(x_k) - m_k(s_k) \;\geq\; \frac{1}{6}\,\sigma_k\,\|s_k\|^3$$

More general **Cauchy decrease bound**:

$$f(x_k) - m_k(s_k) \;\geq\; \frac{\|g_k\|}{6\sqrt{2}}\,\min\left\{\frac{\|g_k\|}{1+\|B_k\|},\; \frac{1}{2}\sqrt{\frac{\|g_k\|}{\sigma_k}}\right\}$$

Key complexity theorem [Nesterov-Polyak 2006; CGT 2011]

Under Lipschitz Hessian (constant L), CR/ARC finds $\|\nabla f(x_k)\| \leq \varepsilon$ in at most

Intuition for the $\varepsilon^{-3/2}$ bound:

On each successful step with $\|g_k\| > \varepsilon$:

$$f(x_k) - f(x_{k+1}) \;\geq\; c\,\varepsilon^{3/2}$$

Since f is bounded below, at most $\mathcal{O}((f_0 - f^*)/\varepsilon^{3/2})$ such steps can occur.

Second-order complexity [CGT Part II]

$$\|\nabla f(x_k)\| \leq \varepsilon_1, \quad \lambda_{\min}(\nabla^2 f(x_k)) \geq -\varepsilon_2^{1/2}$$

$$\Rightarrow\; \mathcal{O}\left(\max\left\{\varepsilon_1^{-3/2},\, \varepsilon_2^{-3}\right\}\right) \text{ iters.}$$

The ARC Algorithm [Cartis, Gould & Toint 2011]

Algorithm 1 Adaptive Cubic Regularisation (ARC)

Init: x_0, $\sigma_0 > 0$, $\eta_1 = 0.1$, $\eta_2 = 0.9$, $\gamma = 2$
for $k = 0, 1, 2, \ldots$ **do**
  Compute $s_k \approx \arg\min_s m_k(s)$ (at least Cauchy decrease)
  Evaluate ratio:

$$\rho_k = \frac{f(x_k) - f(x_k + s_k)}{f(x_k) - m_k(s_k)}$$

  if $\rho_k \geq \eta_1$ **then accept**: $x_{k+1} = x_k + s_k$
  else reject: $x_{k+1} = x_k$
  end if
  Update σ_{k+1}:
  if $\rho_k > \eta_2$ **then** $\sigma_{k+1} \in (0, \sigma_k] \downarrow$
  else if $\rho_k \geq \eta_1$ **then** $\sigma_{k+1} \in [\sigma_k, \gamma\sigma_k]$
  else $\sigma_{k+1} \in [\gamma\sigma_k, \gamma^2\sigma_k] \uparrow$
  end if
end for

How σ_k adapts:

Very successful
$\rho_k > \eta_2$
Model $\approx$ function
$\sigma_{k+1} \leq \sigma_k$ (less cautious)

Unsuccessful
$\rho_k < \eta_1$
Model overpredicts
$\sigma_{k+1} \geq \gamma\sigma_k$
(more cautious)

Successful
$\eta_1 \leq \rho_k \leq \eta_2$
Moderate agreement
$\sigma_{k+1} \in [\sigma_k, \gamma\sigma_k]$

Local Convergence of ARC

Near a **strict local minimizer** x^* (where $\nabla^2 f(x^*) \succ 0$):

Quadratic local convergence [Nesterov-Polyak 2006]

With exact Hessians and Lipschitz $\nabla^2 f$:

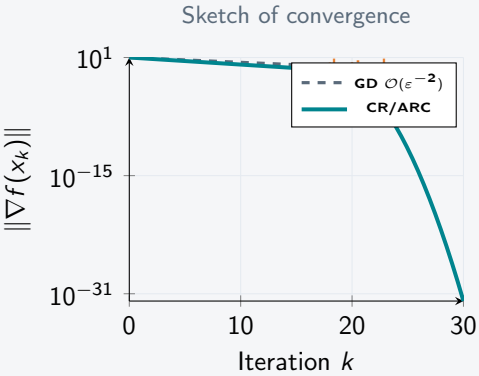
$$\frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|^2} \leq C \quad \text{(bounded)}$$

Superlinear convergence of ARC [CGT Part I, Cor. 4.8]

Under a Dennis-Moré condition on B_k:

$$\|x_{k+1} - x^*\| = o(\|x_k - x^*\|)$$

Three convergence phases:



Schematic only — actual curves depend on problem structure.

Comparison with Other Methods

Method	1st-order compl.	Local conv.	Global guarantee	Key limitation
Gradient Descent	$\mathcal{O}(\varepsilon^{-2})$	Linear	✓	No curvature; slow
Conj. Gradient	$\mathcal{O}(\varepsilon^{-2})$	N/A	✓	First-order only
Newton (plain)	$\mathcal{O}(\varepsilon^{-2})$	Quadratic	×	No global guarantee
L-BFGS	$\mathcal{O}(\varepsilon^{-2})$	Superlinear	×	Approx. Hessian
Trust-Region (std)	$\mathcal{O}(\varepsilon^{-2})$	Superlinear	✓	Hard-constraint subprob.
CRN / ARC	$\mathcal{O}(\varepsilon^{-3/2})$	**Quadratic**	✓	Hessian-vector products

Why ARC beats trust-region: the cubic penalty is a *soft constraint*: it adapts step size anisotropically,

Optimality: Carmon et al. (2019) proved a matching *lower bound*

Solving the Cubic Subproblem

Optimality condition (with $\lambda^* = \sigma \|s^*\|$):

$$(B + \lambda^* I)\, s^* = -g, \quad B + \lambda^* I \succeq 0$$

Reduces to a **1-D secular equation** in λ^*:

$$\pi(\lambda) \;=\; \frac{1}{\|s(\lambda)\|} - \frac{\sigma}{\lambda} = 0$$

Solved by Newton iteration on π.

Approaches by scale

Scale	Method	Cost
Small	Eigendecomp. + secular	$\mathcal{O}(n^3)$
Medium	Cholesky + Newton	$\mathcal{O}(n^3)$
Large	Lanczos Krylov	$\mathcal{O}(n \cdot j)$
Very large	GD on subproblem	$\mathcal{O}(n)$/iter

Lanczos method (large-scale):

Build Krylov subspace
$\mathcal{K}_j(B, g) \;=\; \mathrm{span}\{g, Bg, \ldots, B^{j-1}g\}$

↓

Tridiagonal reduction
$B \;\approx\; V_j T_j V_j^\top$

↓

Solve j-dim cubic exactly
(eigendecomp. of T_j)

↓

Return $s_k \;=\; V_j \tilde{s}_j$

One Hessian-vector product per Lanczos step: practical for $n \sim 10^6$.

Non-Convex Regularizer [Reddi et al. 2016, arXiv:1603.06159]

Standard logistic regression uses an ℓ_2 penalty. We also consider the **non-convex regularizer**:

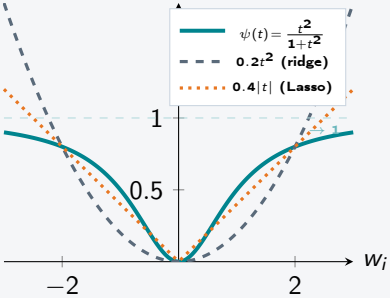
$$r(w) \;=\; \lambda \sum_{i=1}^{d} \frac{\alpha\, w_i^2}{1 + \alpha\, w_i^2}$$

Properties of $\psi(t) = \frac{\alpha t^2}{1+\alpha t^2}$:

- Smooth (differentiable everywhere)
- **Bounded**: $\psi(t) \to 1/\alpha$ as $|t| \to \infty$
- Concave for large $|t|$ — *non-convex!*
- Encourages sparsity via saturating penalty
- Reduces to ridge (ℓ_2^2) for small $|w_i|$

Experiments: $\lambda = 0.001$, $\alpha = 1$.

Penalty comparison



The saturating shape makes the full objective non-convex — ideal for testing CR's ability to handle non-convex problems.

Recent Developments

Stochastic & large-scale variants:

- **Sub-sampled CR** [Kohler & Lucchi 2017]: sub-sampled gradients and Hessians; global convergence preserved
- **SCR** [Tripuraneni et al. 2018]: stochastic cubic regularization, $\tilde{\mathcal{O}}(\varepsilon^{-3.5})$ oracle calls
- **SVRC/SRVRC** [Zhou et al. 2020]: variance-reduced cubic, $\tilde{\mathcal{O}}(d\,\varepsilon^{-3})$
- **SSCN** [Hanzely et al. 2020]: random subspace cubic Newton; interpolates GD $\leftrightarrow$ CR

Riemannian extensions:

- ARC on manifolds [Agarwal et al. 2021]: same $\mathcal{O}(\varepsilon^{-3/2})$ rate on curved spaces

Optimality — lower bounds:

Carmon et al. 2019 (deterministic)

Any p-th order method requires $\Omega(\varepsilon^{-(p+1)/p})$ evaluations. For $p = 2$: $\Omega(\varepsilon^{-3/2})$ — ARC is **worst-case optimal**.

Arjevani et al. 2020 (stochastic)

Stochastic 2nd-order methods need $\Omega(\varepsilon^{-3})$ — an *elbow effect*: higher order cannot improve beyond ε^{-3}.

Higher-order tensor methods: [Nesterov 2021] — p-th order gives $\mathcal{O}(\varepsilon^{-(p+1)/p})$, matching lower bound.

Section 3

Methodology & Preliminary Results

Implementation Plan

Methods to implement (Python):

Primary: CRN and ARC

- **CRN** (Nesterov-Polyak): fixed $\sigma = L/2$, exact or Lanczos subproblem
- **ARC**: adaptive σ_k, Lanczos solver for scalability, Cauchy fallback

Baselines:

- Gradient Descent (GD) with Armijo line-search
- Conjugate Gradient (CG / L-BFGS-B via `scipy`)
- Quasi-Newton: L-BFGS

Reproducibility: 3–5 random initialisations,

Experimental setup:

Dataset	Task
Covtype	Binary (class 1 vs. rest) $n = 581\,012$, $d = 54$
A9a	Binary classification $n = 32\,561$, $d = 123$

Losses:

1. Logistic regression (convex):

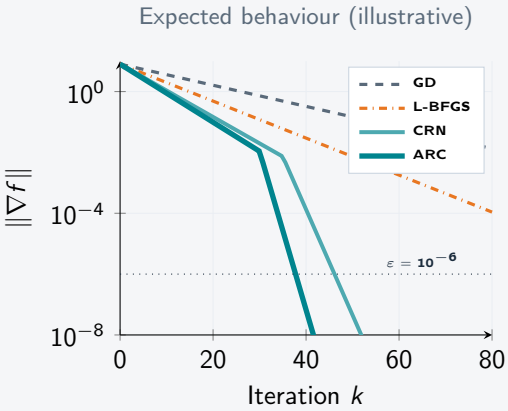
$$\frac{1}{n}\sum_i \log(1 + e^{-y_i w^\top x_i})$$

2. Logistic + non-convex regularizer:

$$+\lambda \sum_j \frac{\alpha w_j^2}{1 + \alpha w_j^2}$$

Anticipated Results

Convergence curves (sketch):



What the experiments will show:

- **Convex loss (logistic):** ARC and CRN expected to reach tight tolerances in fewer iterations than GD/CG, especially once in the quadratic convergence regime.
- **Non-convex loss:** gradient descent and L-BFGS may stall at saddle points; second-order methods better equipped to escape via negative curvature directions.
- **Wall-clock time:** per-iteration cost of CR higher due to Hessian; break-even point with GD at moderate tolerances.
- **Variance across runs:** CR methods

Current status: Theory ✓ | ARC coded (draft) | Experiments pending

Code CRN + ARC in Python; unit tests — *in progress*;
[step,below=0.5cm of imp] (exp) **Experiments**

Run on Covtype + A9a; 3–5 seeds each;
[step,below=0.5cm of thy] (ana) **Analysis**

Compare convergence curves; table of results;
[step,below=0.5cm of lit] (rep) **Final Report**

Write-up, presentation polish;
[arr] (lit)–(thy); [arr] (thy)–(imp); [arr] (imp)–(exp); [arr] (exp)–(ana); [arr] (ana)–(rep);

Summary

Key takeaways so far

1. **Cubic regularization** adds a $\frac{\sigma}{3}\|s\|^3$ penalty to the quadratic Newton model — guaranteeing global descent and handling indefinite Hessians.
2. **CRN** (Nesterov-Polyak 2006) achieves $\mathcal{O}(\varepsilon^{-3/2})$ with fixed $\sigma = L/2$.
3. **ARC** (Cartis-Gould-Toint 2011) adapts σ_k without knowing L, matching the same bound with local **quadratic** convergence.
4. Both methods are **worst-case optimal** among second-order methods (matching lower bound, Carmon et al. 2019).
5. The cubic subproblem is tractable at large scale via **Lanczos-based** Krylov solvers.

Coming up:

- Full Python implementation of ARC with Lanczos subproblem solver
- Benchmark on Covtype and A9a with convex and non-convex losses
- Convergence plots averaged over multiple seeds
- Timing comparison: iterations vs. wall-clock time

Questions?

Thank you

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